How to Kernelize SVM

Example: Polynomial Kernel

$$L_D(\alpha) = \sum_{i=1}\alpha_i - \frac{1}{2}\sum_{i=1}\sum_{j=1}\alpha_i\alpha_j y_i y_j X_i^T X_j$$

$w = \sum_{i=1}\alpha_i y_i X_i$

$$b = -\frac{1}{2}\left[\max_{i:y_i=-1} w^T X_i + \min_{i:y_i=+1} w^T X_i\right]$$

Class of vector $Z \rightarrow \text{sign}(w^T Z + b)$

Kernelize using

$K(a, b) = [a^T b + 1]$

$$L_D(\alpha) = \sum_{i=1}\alpha_i - \frac{1}{2}\sum_{i=1}\sum_{j=1}\alpha_i\alpha_j y_i y_j [X_i^T X_j + 1]^d$$

$w = \sum_{i=1}\alpha_i y_i X_i$

$$b = -\frac{1}{2}\left[\max_{i:y_i=-1} [w^T X_i + 1]^d + \min_{i:y_i=+1} [w^T X_i + 1]^d\right]$$

Class of vector Z is $\text{sign}([w^T Z + 1]^d + b)$